Bundle Structures in Topology

Labix

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Abstract

- Notes on Algebraic Topology by Oscar Randal-Williams

Contents

1 Vector Bundles

1.1 Basic Definitions

Definition 1.1.1 (Local Trivialization Charts)
Let F be a field. Let E, B be spaces. Let $p : E \to B$ be a map. A local trivialization chart of p is a pair (U, ϕ) where $U \subseteq B$ is open and $\phi : p^{-1}(U) \to U \times F^k$ is a homeomorphism such that the following diagram commutes:

$$\begin{tikzcd} p^{-1}(U_i) \arrow[rr, "\phi_i"] \arrow[dr, "p"'] & & U_i \times F^k \arrow[dl, "\pi"] \\ & U_i & \end{tikzcd}$$

Definition 1.1.2 (Vector Bundles)
Let F be a field. Let E, B be spaces with B connected. Let $p : E \to B$ be a continuous map. We say that p is a vector bundle of dimension $d \in \mathbb{N}$ over F if the following are true.

- $p : E \to B$ is surjective.
- There exists a collection $\{(U_i \subseteq B, \phi_i : p^{-1}(U) \to U \times F^d) \mid i \in I\}$ of local trivialization charts such that $B = \bigcup_{i \in I} U_i$.
- For each $b \in B$, the map $\phi_i|_{p^{-1}(b)} : p^{-1}(b) \to \{b\} \times F^k$ is a linear isomorphism.

We say that B is the base space, E the total space. It is denoted as (F, E, B).

The local trivialization means that locally at a neighbourhood, the vector bundle looks the same the open set times F^k. In particular, there is also a notion of trivial bundle which means that the bundle is globally just $B \times \mathbb{R}^r$.

Definition 1.1.3 (The Fiber of a Bundle)
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. Let $b \in B$. Define the fiber of E at B to be
$$E_b = p^{-1}(b)$$

Example 1.1.4 (The Trivial Bundle)
Let B be a space. Let F be a field. The trivial bundle $p : B \times F^k \to B$ defined by $p(b, v) = b$ is a vector bundle. The bundle is denoted by e^k.

Example 1.1.5 (The Mobius Strip)
Let $E = \frac{I \times \mathbb{R}}{\sim}$ where $(0, x) \sim (1, -x)$ for $x \in \mathbb{R}$. Then the projection map $E \to \frac{I}{0 \sim 1} \cong S^1$ is a vector bundle of rank 1.

Example 1.1.6 (The Real Line Bundle over $\mathbb{RP}^n$)
Consider $\mathbb{RP}^n = \frac{\mathbb{R}^{n+1} \setminus \{0\}}{\sim}$ where $x \sim y$ if $y = \lambda x$ for some $\lambda \in \mathbb{R}^\times$. Define the line bundle
$$\gamma_1 : E_{\gamma_1} = \{([x], v) \in \mathbb{RP}^n \times \mathbb{R}^{n+1} \mid v \in [x]\} \to \mathbb{RP}^n$$
to be the map $([x], v) \mapsto [x]$ where $v \in [x]$ or $v = 0$. The local trivialization charts are given by the open cover $U_0, \dots, U_n$ where
$$U_i = \{[x_0 : \cdots : x_n] \mid x_i \neq 0\}$$
and the map $\gamma_1^{-1}(U_i) \to U_i \times \mathbb{R}$ is given by $([x], v) \mapsto ([x], \lambda)$ where $v = \lambda(x_0, \cdots, x_{i-1}, 1, x_{i+1}, \cdots : x_n)$ and $[x] = [x_0 : \cdots : x_{i-1} : 1 : x_{i+1} : \cdots : x_n]$.

Example 1.1.7 (The Complex Line Bundle over $\mathbb{CP}^n$)
Consider $\mathbb{CP}^n = \frac{\mathbb{C}^{n+1} \setminus \{0\}}{\sim}$ where $x \sim y$ if $y = \lambda x$ for some $\lambda \in \mathbb{C}^\times$. Define the line bundle $\gamma_1 : \{([x], v) \in \mathbb{CP}^n \times \mathbb{C}^{n+1} \mid v \in [x]\} \to \mathbb{CP}^n$ to be the map $([x], v) \mapsto [x]$ where $v \in [x]$ or $v = 0$.

Lemma 1.1.8
Let F be a field. Let $p : E \to B$ be a vector bundle of dimension $d \in \mathbb{N}$ over F. Then the local trivialization charts induces a homeomorhpsim

$$E_b \cong F^d$$

for all $b \in B$.

Definition 1.1.9 (Morphism of Vector Bundles)
Let $p_1 : E_1 \to B_1$ and $p_2 : E_2 \to B_2$ be vector bundles. A morphism of these vector bundles is given by is a pair of continuous maps $f : E_1 \to E_2$ and $g : B_1 \to B_2$ such that the following diagram commutes

$$\begin{tikzcd} E_1 \arrow[r, "f"] \arrow[d, "p_1"] & E_2 \arrow[d, "p_2"] \\ B_1 \arrow[r, "g"] & B_2 \end{tikzcd}$$

and $f|_{p^{-1}(b)}$ is a linear map for all $b \in B$.

Definition 1.1.10 (Isomorphism of Vector Bundles) A bundle homomorphism from E_1 to E_2 is an isomorphism if there exists an inverse bundle homomorphism from E_2 to E_1. In this case, we say that E_1 and E_2 are isomorphic.

Lemma 1.1.11
Let B be space. Let $p_1 : E_1 \to B$ and $p_2 : E_2 \to B$ be vector bundles. Let $f : E_1 \to E_2$ be a morphism of vector bundles. Then f is an isomorphism if and only if $f|_{p_1^{-1}(b)}$ is an isomorphism for all $b \in B$.

Proof
If f is an isomorphism, then the inverse of $f|_{p_1^{-1}(b)}$ is clearly given by $f^{-1}|_{p_2^{-1}(b)}$.

Now suppose that $f|_{p_1^{-1}(b)}$ is an isomorphism for all $b \in B$. Then clearly f is bijective. I claim that $p_2 = p_1 \circ f^{-1}$. Let $x \in E_2$. Then there exists $y \in E_1$ such that $f(y) = x$. Then $p_1(f^{-1}(x)) = p_1(y)$. But $p_1 = p_2 \circ f$ by definition of morphisms of vector bundles. Hence $p_1(y) = p_2(f(y)) = p_2(x)$ and we conclude.

It remains to show that f^{-1} is continuous. Let $U \subseteq E_1$ be an open set. Then there exists local trivialization charts (V_i, ϕ_i) of E_1 such that the V_i's cover $p_1(U)$. Then we have $U = \bigcup_{i \in I} U \cap V_i$ and $f^{-1}(U)$ is open if and only if $f^{-1}(U \cap V_i)$ is open. But since restrictions of local trivialization charts are also local trivialization charts, it suffices to prove that f is continuous on local trivialization charts.

So let (U, ϕ) be a local trivialization chart of E_1. Since E_2 is also covered by local trivialization charts, WLOG assume that (U, ψ) is a local trivialization chart of E_2. Now the map $\psi \circ f \circ \phi^{-1} : U \times F^k \to U \times F^n$ sends (x, v) to $(x, g(x)(v))$ for some function $g(x) : F^k \to F^n$. Since the map is a homeomorphism, the map $v \mapsto g(x)(v)$ must be a homeomorphism, and hence $k = n$. Since restriction to each fiber is a bijection, $g(x)$ must be an invertible linear map and hence continuous. Hence $\phi \circ f^{-1} \circ \psi^{-1}$ is continuous. Thus f is continuous. ■

Definition 1.1.12 (Set of Isomorphism Classes of Vector Bundles)
Let X be a space. Let F be a field. Denote the set of isomorphism classes of vector bundles by

$$\text{Vect}_n^F(X) = \frac{\{p : E \to B \mid E \text{ is a vector bundle of } B \text{ over } F\}}{\sim}$$

where $E \sim E'$ if E and E' are isomorphic bundles.

1.2 Sections and Frames

Definition 1.2.1 (Local Sections)
Let k be a field. Let $p : E \to B$ be a vector bundle over k. Let $U \subseteq B$ be an open subset. A local section of p is a map $s : U \to E$ such that $p \circ s = \text{id}_U$.

Lemma 1.2.2
Let k be a field. Let $p : E \to B$ be a vector bundle over k. Let $U \subseteq B$ be an open subset. Then the following are true.

- Let $s_1, s_2 : U \to E$ be local sections of p. Then $s_1 + s_2$ is a section of p.
- Let $s : U \to E$ be a local section of p. Let $\lambda \in k$. Then λs is a section of p.

Definition 1.2.3 (The Space of Sections)
Let F be a field. Let $p : E \to B$ be a vector bundle over F. Let $U \subseteq B$ be an open subset. Define the F-vector space of local sections to be the set

$$s(U, E) = \{s : U \to E \mid s \text{ is a section of } p\}$$

together with addition and scalar multiplication defined pointwise. We write $s(B, E) = s(E)$ as the space of global sections.

Lemma 1.2.4
Let F be a field. Let $p : E \to B$ be a vector bundle over F. Then E is isomorphic to the trivial bundle if and only if there exists global sections $s_1, \dots, s_n \in s(E)$ such that $s_1(b), \dots, s_n(b)$ is a basis for $p^{-1}(b)$ for all $b \in B$.

Example 1.2.5
The Mobius vector bundle $M \to S^1$ is non-trivial.

Proof
Suppose that it is trivial. By the above lemma, there exists a global section $s : S^1 \to M$ such that $s(e^{2\pi it}) \neq [(t, 0)]$ for all $t \in [0, 1]$. Writing $M = \frac{I \times \mathbb{R}}{\sim}$ where $(0, x) \sim (1, -x)$, we see that the map s is of the form $s(e^{2\pi it}) = [(t, f(t))]$ for some continuous map $f : I \to \mathbb{R}$ such that $f(0) = -f(1)$ by definition of the equivalence relation. But by the intermediate value theorem, there exists $c \in I$ such that $f(c) = 0$, contradicting the fact that s is nowhere vanishing. ■

Definition 1.2.6 (Local Frames)
Let F be a field. Let $p : E \to B$ be a vector bundle over F. Let $U \subseteq B$ be an open set. A local frame of E at U is a collection of local sections $s_1, \dots, s_n : U \to E$ such that $s_1(b), \dots, s_n(b)$ is a basis for E_b.

1.3 The Cocycle Conditions

Given two charts (U_α, ϕ_α) and (U_β, ϕ_β) of a vector bundle,

$$\phi_\beta \circ \phi_\alpha^{-1} : (U_\alpha \cap U_\beta) \times F^k \to (U_\alpha \cap U_\beta) \times F^k$$

is a well defined function. In particular, by fixing a point in $U_\alpha \cap U_\beta$, we obtain a linear map.

Definition 1.3.1 (Transition Functions)
Let $p : E \to B$ be an F-vector bundle of rank k. Let (U_α, ϕ_α) and (U_β, ϕ_β) be local trivialization. Define $g_{\alpha,\beta} : U_\alpha \cap U_\beta \to \text{GL}(n, F)$ by

$$x \mapsto \phi_\beta \circ \phi_\alpha^{-1}(x, -) : F^k \to F^k$$

In other words, $g_{\alpha,\beta}$ is such that

$$\phi_\beta \circ \phi_\alpha^{-1}(x, v) = (x, g_{\alpha,\beta}(x)v)$$

For $x \in U_\alpha \cap U_\beta$ and $v \in F^k$.

Proposition 1.3.2
Let $p : E \to B$ be a K-vector bundle of rank k. The transition functions of the vector bundle satisfies the following.
- Cocycle condition: $g_{\gamma,\alpha}(x) \circ g_{\beta,\gamma}(x) \circ g_{\alpha,\beta}(x) = \text{id}$ on $U_\alpha \cap U_\beta \cap U_\gamma$ for all $x \in U_\alpha \cap U_\beta \cap U_\gamma$.
- $g_{\alpha\alpha}(x) = I_r$ for all $x \in U_\alpha$.

Proof
We have
$$\phi_\alpha \circ \phi_\gamma^{-1} \circ \phi_\gamma \circ \phi_\beta^{-1} \circ \phi_\beta \circ \phi_\alpha^{-1} = \text{id}$$
Hence
$$(x, (g_{\gamma,\alpha}(x) \circ g_{\beta,\gamma}(x) \circ g_{\alpha,\beta}(x))(v)) = (x, v)$$
for all $x \in U_\alpha \cap U_\beta \cap U_\gamma$ and all $v \in F^k$. Hence the cocycle condition is achieved.

Clearly $g_{\alpha,\alpha}(x, -)$ is the identity by definition. ∎

The cocycle conditions completely determine a vector bundle.

Definition 1.3.3 (Associated Vector Bundle)
Let F be a field. Let B be a space. Let $\{U_i \mid i \in I\}$ be an open cover of B. Let $g_{i,j} : U_i \cap U_j \to \text{GL}(k, F)$ be a continuous functions for $i, j \in I$ such that $g_{k,i}(x) \circ g_{j,k}(x) \circ g_{i,j}(x) = \text{id}$ for all $x \in U_i \cap U_j \cap U_k$ and $g_{i,i}(x) = \text{id}$ for all $x \in U_i$. Define the associated vector bundle to be the space
$$E_g = \frac{\coprod_{i \in I} U_i \times F^k}{\sim}$$
where $(i, b, v) \sim (j, b, g_{i,j}(b)(v))$ if $b \in U_i \cap U_j$, together with the induced projection map $\text{proj} : E_g \to B$.

Lemma 1.3.4
Let F be a field. Let B be a space. Let $\{U_i \mid i \in I\}$ be an open cover of B. Let $g_{i,j} : U_i \cap U_j \to \text{GL}(k, F)$ be a continuous functions for $i, j \in I$. Then $p : E_g \to B$ is a vector bundle of rank k.

Proof
Firstly note that the projection map $E_g \to B$ is well defined by the universal property of quotient. We then need to prove that the local triviality condition is satisfied. Clearly the composition $U_i \times F^k \hookrightarrow \coprod_{i \in I} U_i \times F^k \to E_g$ composed with the canonical projection map commutes. It is a homeomorphism following the universal property of quotients. ∎

Lemma 1.3.5
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. Let $g = \{g_{i,j} \mid i, j \in I\}$ be the transition functions. Then the transition functions induce an isomorphism
$$E_g \cong E$$
of vector bundles.

1.4 Operations on Vector Bundles

Definition 1.4.1 (Whitney Sum) Let $p_1 : E_1 \to B$ and $p_2 : E_2 \to B$ be two vector bundles. Define the direct sum of the vector bundles to be

$$E_1 \oplus E_2 = \{(v_1, v_2) \in E_1 \times E_2 \mid p_1(v_1) = p_2(v_2)\}$$

together with the projection $p : E_1 \oplus E_2 \to B$ defined by $(v_1, v_2) \mapsto p_1(v) = p_2(v)$.

Example 1.4.2
Let $p : M \to S^1$ be the Mobius vector bundle. Then $M \oplus M \cong e^2$.

Proof
Define a morphism $M \oplus M \to e^2$ by $([(t, x)], [(t, y)]) \mapsto (e^{2\pi it}, x, y)$. Notice that that restricted to the open cover $U_1 = \{e^{2\pi it} \mid 0 \leq t \leq 1/2\}$ and $U_2 = \{e^{2\pi it} \mid 1/2 \leq t \leq 1\}$ of S^1. we see that the local trivialization charts of $M \oplus M$ coincides with that of our morphism. Hence the morphism is a well defined morphism of vector bundles. Moreover, it is an isomorphism of vector spaces on each fiber. ■

Definition 1.4.3 (Pullback Bundle)
Let F be a field. Let X, Y be spaces. Let $p : E \to Y$ be a vector bundle over F. Let $f : X \to Y$ be a map. Define the pullback bundle of f to be the space

$$f^*(E) = \{(x, e) \in X \times E \mid f(x) = p(e)\}$$

together with the projection map $p^* : f^*(E) \to X$ given by $p^*(x, e) = x$.

Definition 1.4.4 (The Hom Bundle)
Let F be a field. Let B be a space. Let $p_1 : E_1 \to B$ and $p_2 : E_2 \to B$ be vector bundles over F. Define the Hom bundle of E_1 and E_2 to be the space

$$\text{Hom}(E_1, E_2) = \coprod_{b \in B} \text{Hom}_F(p_1^{-1}(b), p_2^{-1}(b))$$

together with the coproduct of the maps $\text{Hom}_F(p_1^{-1}(b), p_2^{-1}(b)) \to \{b\}$.

Definition 1.4.5 (The Dual Bundle)
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. Define the dual bundle of p to be the vector bundle

$$E^* = \text{Hom}(E, e^1)$$

Definition 1.4.6 (The Tensor Product Bundle)
Let F be a field. Let B be a space. Let $p_1 : E_1 \to B$ and $p_2 : E_2 \to B$ be vector bundles over F. Define the tensor product bundle to be the space

$$E_1 \otimes E_2 = \coprod_{b \in B} p_1^{-1}(b) \otimes_F p_2^{-1}(b)$$

together with the coproduct of the maps $p_1^{-1}(b) \otimes p_2^{-1}(b) \to \{b\}$.

Definition 1.4.7 (Tensor Product with a Vector Space)
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. Let V be an F-vector space. Define the tensor product of E and V to be

$$E \otimes V = E \otimes (B \times V)$$

Definition 1.4.8 (The Exterior Product Bundle)
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle. Define the nth exterior

product bundle of E to be

$$\Lambda^n(E) = \coprod_{b \in B} \Lambda^n(p^{-1}(b))$$

together with the coproduct of the maps $\Lambda^n(p_1^{-1}(b)) \to \{b\}$.

1.5 Homotopy Theory of Vector Bundles

Theorem 1.5.1 (The Homotopy Theorem)
Let X be a paracompact space. Let $p : E \to X \times I$ be a vector bundle. Then there is a bundle isomorphism

$$p^{-1}(X \times \{0\}) \cong p^{-1}(X \times \{1\})$$

Proposition 1.5.2
Let X, Y be paracompact spaces. Let $p : E \to Y$ be a vector bundle. Let $f, g : X \to Y$ be homotopic maps. Then there is an isomrophism

$$f^*(E) \cong g^*(E)$$

Corollary 1.5.3
Let X, Y be paracompact spaces. Let $f : X \to Y$ be a map. If f is a homotopy equivalence, then the pullback map induces natural bijection

$$f^* : \text{Vect}_n^F(Y) \stackrel{1:1}{\to} \text{Vect}_n^F(X)$$

Corollary 1.5.4
Let B be a paracompact space. Let $p : E \to B$ be a vector bundle. If B is contractible, then E is trivial.

2 Sequences of Vector Bundles

2.1 Subbundles and Quotient Bundles

Definition 2.1.1 (Vector Subbundles)
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. A subbundle of E is a subset $E' \subseteq E$ such that $p|_{E'} : E' \to B$ is a vector bundle over F.

Lemma 2.1.2
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. Let E' be a subbundle of E. Let $\phi_i : p^{-1}(U_i) \to U_i \times F^k$ be a local chart of E'. Then the following are true.

- There exists a map $\psi_i : p^{-1}(U_i) \to U_i \times F^{n-k}$ such that the map $p^{-1}(U_i) \to U_i \times F^n$ given by $x \mapsto (x, \text{proj}_{F^k}(\phi_i(x)), \text{proj}_{F^{n-k}}(\psi_i(x)))$ is a local chart of E.
- The transition functions of E are given in block matrix
$$g_{ij} = \begin{pmatrix} g'_{ij} & l_{ij} \\ 0 & k_{ij} \end{pmatrix}$$
for some maps $l_{ij} : U_i \cap U_j \to M_{k,n-k}(F)$ and $k_{ij} : U_i \cap U_j \to \text{GL}(n-k, F)$.
- The maps k_{ij} satisfy the cocycle condition and $k_{ii}(x) = I_{n-k}$ for all $x \in U_i$.

Definition 2.1.3 (The Quotient Bundle)
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. Let E' be a subbundle of E. Define the quotient bundle E/E' to be the associated vector bundle to the collection of functions k_{ij} given by the above.

Lemma 2.1.4
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. Let E' be a subbundle of E. There are linear isomorphisms
$$(E/E')_b \cong E_b/E'_b$$
for all $b \in B$.

Proposition 2.1.5 (Universal Property of Quotient Bundles)
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. Let E' be a subbundle of E. Then the quotient bundle and the quotient map $q : E \to E/E'$ satisfies the following universal property. For any vector bundle G over F and any morphism $f : E \to G$ such that $f(E')$ is the trivial bundle, there exists a unique morphism $\widetilde{f} : E/E' \to G$ such that the following diagram commutes:
$$\begin{tikzcd} E \arrow[r, "q"] \arrow[dr, "f"'] & E/E' \arrow[d, "\widetilde{f}"] \\ & G \end{tikzcd}$$
Moreover, E/E' is the unique (up to unique isomorphism) vector bundle that satisfies the above.

2.2 The Kernel of a Morphism

2.3 Short Exact Sequences of Vector Bundles

Definition 2.3.1 (Short Exact Sequences of Vector Bundles)
Let F be a field. Let B be a space. Let E, G, H be vector bundles over B. Let $f : E \to G$ and $g : G \to H$ be morphisms. We say that the sequence
$$0 \longrightarrow E \xrightarrow{f} G \xrightarrow{g} H \longrightarrow 0$$
is a short exact sequence of vector bundles if the induced linear maps

$$0 \longrightarrow E_b \xrightarrow{f|_{E_b}} G_b \xrightarrow{g|_{E_b}} H_b \longrightarrow 0$$

are short exact sequences of vector spaces for all $b \in B$.

Lemma 2.3.2
Let F be a field. Let B be a space. Suppose that the following is a short exact sequence of vector bundles over B:

$$0 \longrightarrow E \xrightarrow{f} G \xrightarrow{g} H \longrightarrow 0$$

Then the following are true.
- g induces an isomorphism $G/E \cong H$.
- f induces an isomorphism $\ker(G \to H) \cong E$.

TBA: Exactness of taking sections TBA: Exactness of tensor products

Note: category of vector bundles is not an abelian category.

3 Reduction of Structure Group

3.1 Structure Groups

Definition 3.1.1 (Reduction of Structure Groups)
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle over F. Let $H \leq \mathrm{GL}(n, F)$ be a subgroup. We say that E admits a reduction of structure groups to H if there exists a vector bundle $p : E' \to B$ over F such that the following are true.

- The transition functions $g_{\alpha,\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}(n, F)$ satisfy $\mathrm{im}(g_{\alpha,\beta}) \leq H$ for any local trivialization charts (U_α, ϕ_α) and (U_β, ϕ_β).
- E and E' are isomorphic vector bundles.

3.2 Metrics Structures

Definition 3.2.1 (Metrics)
Let F be a field. Let B be a space. Let $p : E \to B$ be a vector bundle.

- Suppose that $F = \mathbb{R}$. A metric on E is a smooth section of $s \in (E \otimes E)^* \cong E^* \otimes E^*$ such that $s_p : E_b \times E_b \to \mathbb{R}$ is an inner product for all $b \in B$.
- Suppose that $F = \mathbb{C}$. A metric on E is a smooth section of $s \in (E \otimes E)^* \cong E^* \otimes E^*$ such that $s_p : E_b \times E_b \to \mathbb{R}$ is a Hermitian product for all $b \in B$.

Lemma 3.2.2
Let $F = \mathbb{R}$ or $\mathbb{C}$. Let B be a space. Let $p : E \to B$ be a vector bundle. If B is paracompact, then E admits a metric.

Note: ses of vector bundles of paracompact spaces are split exact. Note: vector bundles have reduction of structure group to $O(n)$ and $U(n)$ if and only if they admit a metric.

3.3 Orientability

Definition 3.3.1 (Orientation on Vector Spaces)
Let V be a finite dimensional vector space over $\mathbb{R}$. An orientation of V is an equivalence class $[B] \in \frac{\{B \subseteq V \mid B \text{ is a basis for } V\}}{\sim}$ where $B \sim C$ if the change of basis matrix $[T]_B^C$ satisfies $\det\left([T]_B^C\right) > 0$.

Lemma 3.3.2
Let V be a finite dimensional vector space over $\mathbb{R}$. Then V admits exactly two orientations.

Proposition 3.3.3
Let V, W be finite dimensional vector spaces over $\mathbb{R}$. Let $T : V \to W$ be a linear isomorphism. If $[B]$ is an orientation of V, then $[T(B)]$ is an orientation of W.

Definition 3.3.4 (Orientation of a Vector Bundle)
Let B be a space. Let $p : E \to B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Let w_b be an orientation of E_b for each $b \in B$. We say that $(w_b)_{b \in B}$ is an orientation of E if for each local trivialization chart (U, ϕ) and any chosen orientation u of $\mathbb{R}^n$, the map $U \to \{\pm 1\}$ given by

$$b \mapsto \begin{cases} 1 & \text{if } \phi_{E_b}(w_b) = u \\ -1 & \text{otherwise} \end{cases}$$

is a constant map.

Definition 3.3.5 (Orientability)
Let B be a space. Let $p : E \to B$ be a vector bundle of rank n with fiber $\mathbb{R}$. We say that E is orientable if E admits an orientation.

Proposition 3.3.6
Let B be a space. Let $p : E \to B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Then E is orientable if and only if E admits a reduction of structure group to $\mathrm{SL}(n, \mathbb{R})$.

Frame bundle Fr(E)

orientable iff Fr(E) times +-1 over general linear group admits global section

3.4 Subbundles

$O(n)$-vector bundle admits k-dimensional subbundle if and only if it has reduction of structure group to $O(k) \times O(n-k)$.

3.5 Almost Complex Structures

Definition 3.5.1 (Linear Complex Structure)
Let V be a vector space over $\mathbb{R}$. A linear complex structure on V is a linear map $J : V \to V$ such that $J^2 = -\mathrm{id}_V$.

Proposition 3.5.2
Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V. Then V is a $\mathbb{C}$-vector space whose action is given by
$$(x + iy) \cdot v = xv + yJ(v)$$
for $v \in V$ and $x + iy \in \mathbb{C}$.

Proposition 3.5.3
Let V be a finite dimensional vector space over $\mathbb{R}$. Then the following are true.
- V admits a linear complex structure if and only if $\dim_{\mathbb{R}}(V)$ is even.
- In this case, we have $\dim_{\mathbb{C}}(V) = 2\dim_{\mathbb{R}}(V)$.

Example 3.5.4
Let $J : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ be the map defined in block matrix form
$$J = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$$
Then J defines a linear complex structure on $\mathbb{R}^{2n}$. In particular, if $e_1, \dots, e_{2n}$ are the standard basis vectors for $\mathbb{R}^{2n}$, then $\frac{1}{2}(e_1 - ie_{2n+1}), \dots, \frac{1}{2}(e_n - ie_{2n})$ is a basis for V as a $\mathbb{C}$-vector space.

Proposition 3.5.5
Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V. Let $T : V \to V$ be an $\mathbb{R}$-linear transformation. Then T is a $\mathbb{C}$-linear transformation if and only if $J \circ T = T \circ J$.

The linear complex structure on an $\mathbb{R}$-vector space gives an intrinsic way of giving the $\mathbb{R}$-vector space the structure of a $\mathbb{C}$-vector space. However, we already know an external way of giving $\mathbb{R}$-vector spaces $\mathbb{C}$-multiplication: namely by extension of scalars.

Proposition 3.5.6
Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V. Then the following are true.
- J extends to a linear complex structure on $V \otimes_{\mathbb{R}} \mathbb{C}$ by $J(v \otimes z) = J(v) \otimes z$.
- The eigenvalues of $J : V \otimes_{\mathbb{R}} \mathbb{C} \to V \otimes_{\mathbb{R}} \mathbb{C}$ is given by $\pm i$ and the eigenspaces of J are given by
$$E_i = \{v \otimes 1 - J(v) \otimes i \mid v \in V\} \quad \text{and} \quad E_{-i} = \{v \otimes 1 + J(v) \otimes i \mid v \in V\}$$
- There is a natural isomorphism of $\mathbb{C}$-vector space $V \cong E_i$ given by $v \mapsto v \otimes 1 - J(v) \otimes i$.

- The conjugation map $\overline{(-)} : V \otimes_R \mathbb{C} \to V \otimes_{\mathbb{R}} \mathbb{C}$ given by $\overline{v \otimes z} = v \otimes \overline{z}$ restricts to an isomorphism
$$\overline{(-)}|_{E_i} : E_i \overset{\cong}{\to} E_{-i}$$
whose inverse is given by $\overline{(-)}|_{E_{-i}}$.

Proposition 3.5.7
Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V. Then J^T defines a linear complex structure on V^*.

Definition 3.5.8 (Almost Complex Structure)
Let B be a space. Let $p : E \to B$ be a vector bundle over $\mathbb{R}$. An almost complex structure on p is a vector bundle isomorphism $j : E \to E$ such that $j \circ j = \text{id}$.

Proposition 3.5.9
Let B be a space. Let $p : E \to B$ be a vector bundle over $\mathbb{R}$. Let J be an almost complex structure on p. Then the following are true.
- The rank of E over $\mathbb{R}$ is even.
- E is a complex vector bundle whose action is given by
$$(x + iy) \cdot v = xv + yJ_b(v)$$
for $v \in E_b$ and $x + iy \in \mathbb{C}$ for each $b \in B$.

Proposition 3.5.10
Let B be a space. Let $p : E \to B$ be a vector bundle over $\mathbb{C}$ of rank n. Then E is a real vector bundle of rank $2n$.

3.6 Symplectic Structures

4 The Topology of Fiber Bundles

4.1 Fiber Bundles

Fiber bundles serve as somewhat of a generalization of both vector bundles and covering spaces, while being a special case of a fibration. It therefore has the properties of a fibration.

Definition 4.1.1 (Fiber Bundles)
Let E, B, F be spaces with B connected, and $p : E \to B$ a continuous map. We say that p is a fiber bundle over F if the following are true.
- $p : E \to B$ is surjective
- Local Triviality: There exists a collection

$$\{(U_k \subseteq B, \phi_k : p^{-1}(U_k) \overset{\cong}{\to} U_k \times F) \mid k \in I\}$$

called local trivialization charts such that the following diagram commutes:

$$\begin{tikzcd} p^{-1}(U_k) \arrow[rr, "\phi_k"] \arrow[dr, "p"'] & & U_k \times F \arrow[dl, "\pi"] \\ & U_k & \end{tikzcd}$$

where π is the projection by forgetting the second variable.

We say that B is the base space, E the total space. It is denoted as (F, E, B) or $F \hookrightarrow E \to B$.

Intuitively, we would like a fiber bundle to locally look like the product $B \times F$.

Lemma 4.1.2
Let K be a field. Let $p : E \to B$ be a fiber bundle with fiber F. Then the local trivialization charts induces a homeomorhpsim

$$p^{-1}(b) \cong F$$

for all $b \in B$.

Fiber bundles generalizes vector bundles in the sense that the fibers are no longer vector spaces but instead arbitrary spaces.

Lemma 4.1.3
The following are true regarding fiber bundles.
- Every vector bundle is a fiber bundle.
- Every covering space is a fiber bundle.

Proof Indeed if $p : E \to B$ is a vector bundle, then each fiber $p^{-1}(b)$ is an n-dimensional vector spaces over a field F. Moreover, by definition the local triviality condition is also satisfied.

Finally, for any $U \subseteq X$, recall that

$$p^{-1}(U) = \coprod_{i \in I} V_i$$

where each $V_i \cong U$. It is clear by definition that $|p^{-1}(x)||I|$ for any $x \in X$. By giving I the discrete topology, we obtain a homeomorphism $p^{-1}(x) \cong I$. The homeomorphism $p^{-1}(U) = \coprod_{i \in I} V_i$ translates to

$$p^{-1}(U) = \coprod_{i \in I} V_i \cong \coprod_{i \in I} U \cong U \times I$$

defined by $\tilde{x} \in V_i \mapsto (p(\tilde{x}) = x, i)$. It is thus clear that the local triviality condition is satisfied. ■

Example 4.1.4
The following maps are fiber bundles.
- Let $f_n : S^1 \to S^1$ be the map defined by $f_n(z) = z^n$. Then f_n is a fiber bundle with fiber the

nth roots of unity.

- Let $f : \mathbb{R} \to S^1$ be the map defined by $f(t) = e^{2\pi it}$. Then f is a fiber bundle with fiber $\mathbb{Z}$.
- Let $p : S^n \to \mathbb{RP}^n$ be the projection map. Then p is a fiber bundle with fiber as a two point set with the discrete topology.

Proof
Indeed, all of these are covering spaces and hence fiber bundles by the above lemma. ∎

Example 4.1.5
Let $M = \frac{[-1,1]\times[-1,1]}{(t,-1)\sim(1-t,1)}$ be the Mobius strip. Then the projection map $p : M \to C$ defined by $p(t,s) = s$ is a fiber bundle with fiber $[-1,1]$.

Example 4.1.6 (The Mapping Torus)
Let X be a space. Let $f : X \to X$ be a homeomorphism. Define the map

$$p : \frac{X \times I}{(x,0) \sim (f(x),1)} \to S^1$$

by $[(x,t)] \mapsto e^{2\pi it}$. Then it is a fiber bundle with fiber X.

Proof
Consider the open cover $U = S^1 \setminus \{1\}$ and $V = S^1 \setminus \{-1\}$. Then $p^{-1}(U) = X \times (0,1)$ and so the local trivialization condition is satisfied. Now $p^{-1}(V) = \frac{X\times[0,1/2)\cup(1/2,1]}{(x,0)\sim(f(x),1)}$ and is clearly homeomorphic to $V \times X$. ∎

Proposition 4.1.7 Every fiber bundle is a Serre fibration.

We can provide a partial converse for the fact that every fiber bundle is a Serre fibration.

Proposition 4.1.8 Let $p : E \to B$ be a fiber bundle. If B is paracompact, then p is a (Hurewicz) fibration.

We there fore have inclusions

$$\substack{\text{Fiber}\\\text{Bundles}} \subset \substack{\text{Serre}\\\text{Fibrations}} \subset \substack{\text{(Hurewicz)}\\\text{Fibrations}}$$

Definition 4.1.9 (Map of Fiber Bundles)
Let (F_1, E_1, B_1) and (F_2, E_2, B_2) be fiber bundles. A map of fiber bundles is a pair of basepoint preserving continuous maps $(\tilde{f} : E_1 \to E_2, f : B_1 \to B_2)$ such that the following diagram commutes:

$$\begin{tikzcd}
E_1 \arrow[r, "\tilde{f}"] \arrow[d, "p_1"'] & E_2 \arrow[d, "p_2"] \\
B_1 \arrow[r, "f"'] & B_2
\end{tikzcd}$$

A map of fiber bundles $(\tilde{f}, f)$ is said to be an isomorphism if there is a map $(\tilde{g} : E_2 \to E_1, g : B_2 \to B_1)$ such that $\tilde{g}$ is the inverse of $\tilde{f}$ and g is the inverse of f.

Such a map of fiber bundles determine a continuous of the fibers $F_1 \cong p_1^{-1}(b_1) \to p_2^{-1}(b_2) \cong F_2$. Moreover, such a morphism preserves fibers. Indeed, If $p_1^{-1}(b)$ is a fiber of B, then using the commutativity of the diagram we have that

$$p_2(\overline{f}(p_1^{-1}(b))) = f(p_1(p_1^{-1}(b))) = f(b)$$

which implies that

$$p_2^{-1}(f(b)) = \overline{f}(p^{-1}(b))$$

or in other words, the fiber at $f(b)$ is the same as the fiber at b applied with $\overline{f}$.

Definition 4.1.10 (Isomorphic Fiber Bundles)
Let $p_1 : E_1 \to B_1$ and $p_2 : E_2 \to B_2$ be two fiber bundles. Let $(\widetilde{f}, f) : p_1 \to p_2$ be a map of fiber bundles. We say that $(\widetilde{f}, f)$ is an isomorphism if there exists a map $(\widetilde{g}, g) : p_2 \to p_1$ of fiber bundles such that $\widetilde{f}$ is the inverse of $\widetilde{g}$ and f is the inverse of g. In this case, we say that p_1 and p_2 are isomorphic.

There are two important special cases of fiber bundles that will appear time and time again.

Definition 4.1.11 (Trivial Bundles)
We say that a fiber bundle (F, E, B) is trivial if (F, E, B) is isomorphic to the trivial fibration $B \times F \to B$.

Definition 4.1.12 (The Pullback Bundle) Let $p : E \to B$ be a fiber bundle with fiber F. Let $f : B' \to B$ be a continuous function. Define the pullback of p by f to be the space

$$f^*(E) = \{(b', e) \in B' \times E \mid p(e) = f(b')\}$$

Theorem 4.1.13 Let $p : E \to B$ be a fiber bundle. Suppose that $f, g : X \to B$ are homotopic maps. Then the pull back bundles

$$f^*(E) \cong g^*(E)$$

are equivalent.

4.2 Sections of a Bundle

Definition 4.2.1 (Sections) Let (F, E, B) be a fiber bundle. A section on the fiber bundle is a map $s : B \to E$ such that

$$p \circ s = \text{id}_B$$

Definition 4.2.2 (Local Sections) Let (F, E, B) be a fiber bundle. Let $U \subset B$ be an open set. A local section of the fiber bundle on U is a map $s : U \to B$ such that

$$p \circ s = \text{id}_U$$

4.3 Sphere Bundles

We now consider a special type of fibrations where the fibers are given by S^1. We define a class of fiber bundles of the form $S^1 \to S^{2n+1} \to \mathbb{CP}^n$. When $n = 1$, this is the Hopf fibration.

Definition 4.3.1 (Sphere Bundles) A sphere bundle is a fiber bundle $p : E \to B$ for which its fibers are the n-sphere S^n.

Example 4.3.2
Let $n \in \mathbb{N}$. Consider S^{2n+1} lying inside $\mathbb{C}^{n+1}$. The projection map $\mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{CP}^n$ induces a fiber bundle

$$S^1 \hookrightarrow S^{2n+1} \to \mathbb{CP}^n$$

Proof
Clearly the projection map $p : S^{2n+1} \to \mathbb{CP}^n$ is surjective. Consider the open cover $\{U_i \mid 0 \le i \le n\}$ where $U_i = \{[z_0 : \cdots : z_n] \mid z_i \ne 0\}$. Define the local trivialization charts $\phi_i : p^{-1}(U_i) \to U_i \times S^1$ by $(z_0, \dots, z_n) \to ([z_0 : \cdots : z_n], z_i/|z_i|)$. It is continuous since it is the product of the composition map and the projection map to the sphere S^1. Its inverse is the continuous map $([z_0, \dots, z_n], w) = w|z_i|z_i^{-1}(z_0, \dots, z_n)$ since it is induced by the universal property of quotients. It is clear that the required diagram commutes, that is $p = \text{proj} \circ \phi_i$ where $\text{proj} : U_i \times S^1 \to U_i$ is the projection map to the first coordinate. ∎

Example 4.3.3
For each $n \in \mathbb{N}$, we have a map of fiber bundles:

$$\begin{tikzcd} S^{2n+1} \arrow[r, hook] \arrow[d] & S^{2n+3} \arrow[d] \\ \mathbb{CP}^n \arrow[r, hook] & \mathbb{CP}^{n+1} \end{tikzcd}$$

whose horizontal maps are induced by the inclusion $\mathbb{C}^n \hookrightarrow \mathbb{C}^{n+1}$ into the first n coordinates. Taking colimit of the diagram, we have that

$$S^1 \hookrightarrow S^\infty \to \mathbb{CP}^\infty$$

is a fiber bundle.

Proof
Since the local triviality condition is a local condition, for each $x \in \mathbb{CP}^\infty$, x lives in $\mathbb{CP}^n$ for some $n \in \mathbb{N}$ and since $S^1 \hookrightarrow S^{2n+1} \to \mathbb{CP}^n$ is a fiber bundle, there exists a chart for which the local triviality condition is satisfied. ■

Example 4.3.4 (Hopf Fibration / Hopf Bundle)
Consider the above example with $n = 1$. Restricting the total space to S^3, the fiber bundle $p : S^3 \to S^2$ with fiber S^1 is called the Hopf fibration / Hopf bundle.